Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Coolahan, James E Ph.D.		
Project Title:	Distributed Simulation of Integrated Human Function		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Cardiovascular Alterations Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Laurel	**State:**	MD
Zip Code:	20723-6099	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	12/01/2000	**End Date:**	04/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Winslow, Raimond (Johns Hopkins University School of Medicine) Feldman, Andrew (Johns Hopkins University Applied Physics Laboratory)		
Grant/Contract No.:	CA.002.15 IHF.002.03		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A synopsis of the four specific aims of the three-year project is as follows: 1. To develop, at JHU, an experimentally-based computational model of the human ventricular myocyte using cells isolated from tissue biopsies performed in patients; and to develop a finite-element model of the geometry and fiber structure of the human heart using diffusion-tensor imaging data to be collected at JHU, fit by a finite-element model to be developed at the University of California, San Diego (UCSD). 2. To develop a distributed simulation of human cardiac function, incorporating a simulation of the human cardiac ventricular cell resident at JHU based on the model discussed above and a simulation of coupled cardiac mechanical and electrical function resident at UCSD, with distributed simulation control based at JHU/APL; 3. Working with the NSBRI team, to select other appropriate cardiovascular system models that can be represented over time using simulations, and integrate them into a distributed simulation of cardiovascular		

	function; and 4. Again working with the NSBRI team, to select other system (e.g., muscle) models that can be represented over time using simulations, and integrate them into a multi-function distributed simulation representative of the full IHF simulations that will be needed for long-duration spaceflight. The project is leveraging the significant body of research in constructing distributed interoperable simulations that has been performed in connection with the development of the High Level Architecture (HLA) for simulations developed by the U. S. Department of Defense.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The long-term objective of the Distributed Simulation of Integrated Human Function project for the National Space Biomedical Research Institute (NSBRI) is to demonstrate the ability to simulate Integrated Human Function (IHF) over time by providing a technical framework to permit simulations of different human physiological functions, executing in separate locations, to interact to produce synergistic results. The hypothesis of this research is that interoperable simulations of human physiological functions applicable to the spaceflight environment executing interactively can produce integrated results that cannot be produced by these simulations executing independently.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)